

DATA ANALYTICS · PROJECT REPORT

VENDORS' PERFORMANCE ANALYSIS

*An end-to-end analysis of vendor sales, profitability
and inventory performance*

SQL · PYTHON (EDA) · POWER BI

REPORT SNAPSHOT

Scope	Vendor, purchase, sales, and inventory performance across the consolidated vendor_sales_summary dataset
Analysis Tools	SQL (data preparation) · Python – pandas / seaborn / statistical testing (EDA) · Power BI (dashboarding)
Coverage	116 vendors · ~6,000 brands · four underlying source tables merged into one analytical table

INSIDE THIS REPORT

Contents

01 Business Problem

02 Data Overview

03 Exploratory Data Analysis — Insights

3.1 Statistical Summary

3.2 Data Filtering

3.3 Correlation Insights

3.4 Research Questions & Key Findings (10 sub-analyses)

04 Power BI Dashboard Overview

4.1 Primary KPIs

4.2 Dashboard — Page 1

4.3 Dashboard — Page 2

05 Final Recommendations

SECTION 01

Business Problem

The challenge driving this analysis

Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- Identify underperforming brands that require promotional or pricing adjustments.
- Determine top vendors contributing to sales and gross profit.
- Analyze the impact of bulk purchasing on unit costs.
- Assess inventory turnover to reduce holding costs and improve efficiency.
- Investigate the profitability variance between high-performing and low-performing vendors.

SECTION 02

Data Overview

Where the numbers in this report come from

The analysis draws on a single aggregated table, **vendor_sales_summary**, which combines every vendor and business-related dataset used in this study. The four underlying sources are summarised below.

DATASET	DESCRIPTION
purchase_prices	Purchase prices of different products and brands across vendors.
purchases	Products purchased from each vendor, including quantities, prices, and transaction details.
vendor_invoice	Vendor invoices — purchase orders, invoice dates, quantities, purchase amounts, freight costs, and payment details.
sales	Sales of vendor-supplied products, including quantities, amounts, selling prices, dates, and related product/vendor details.

All four sources are stored in a database for easy retrieval, and are merged into the single summary table that every analysis in this report is built on.

SECTION 03

Exploratory Data Analysis

Distributions, data quality checks, and the relationships between key metrics

3.1 Statistical Summary

A first pass over the combined dataset surfaces its overall shape, scale, and the data-quality issues that need to be handled before any vendor-level conclusions can be trusted.

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.00	10650.65	18753.52	2.00	3951.00	7153.00	9552.00	201359.00
Brand	10692.00	18039.23	12662.19	58.00	5793.50	18761.50	25514.25	90631.00
Volume	10692.00	847.36	664.31	50.00	750.00	750.00	750.00	20000.00
ActualPrice	10692.00	35.64	148.25	0.49	10.99	15.99	28.99	7499.99
PurchasePrice	10692.00	24.39	109.27	0.36	6.84	10.46	19.48	5681.81
TotalPurchaseQuantity	10692.00	3140.89	11095.09	1.00	36.00	262.00	1975.75	337660.00
TotalPurchaseDollars	10692.00	30106.69	123067.80	0.71	453.46	3655.47	20738.25	3811251.60
TotalSalesAvgPrice	10692.00	33.23	125.26	0.00	10.16	15.15	27.66	5799.99
TotalSalesQuantity	10692.00	3077.48	10952.85	0.00	33.00	261.00	1929.25	334939.00
TotalSalesDollars	10692.00	42239.07	167655.27	0.00	729.22	5298.05	28396.92	5101919.51
TotalExciseTax	10692.00	1774.23	10975.58	0.00	4.80	46.57	418.65	368242.80
FreightCost	10692.00	61433.76	60938.46	0.09	14069.87	50293.62	79528.99	257032.07
GrossProfit	10692.00	12132.38	46224.34	-52002.78	52.92	1399.64	8660.20	1290667.91
ProfitMargin	10692.00	-inf	NaN	-inf	13.32	30.41	39.96	99.72
StockTurnover	10692.00	1.71	6.02	0.00	0.81	0.98	1.04	274.50
SalestoPurchaseRatio	10692.00	2.50	8.46	0.00	1.15	1.44	1.67	352.93

FIGURE 1 Descriptive statistics (count, mean, std, min, quartiles, max) for every numeric field in vendor_sales_summary.

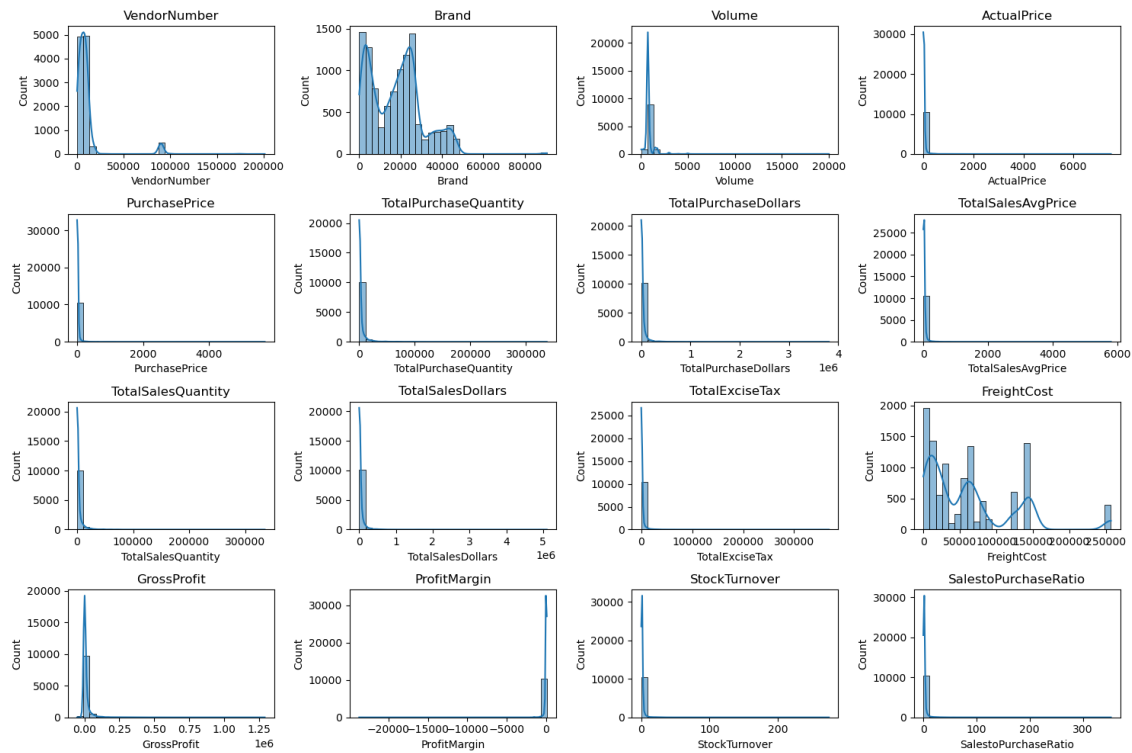


FIGURE 2 Distribution of each numeric variable — most purchase and sales metrics are heavily right-skewed.

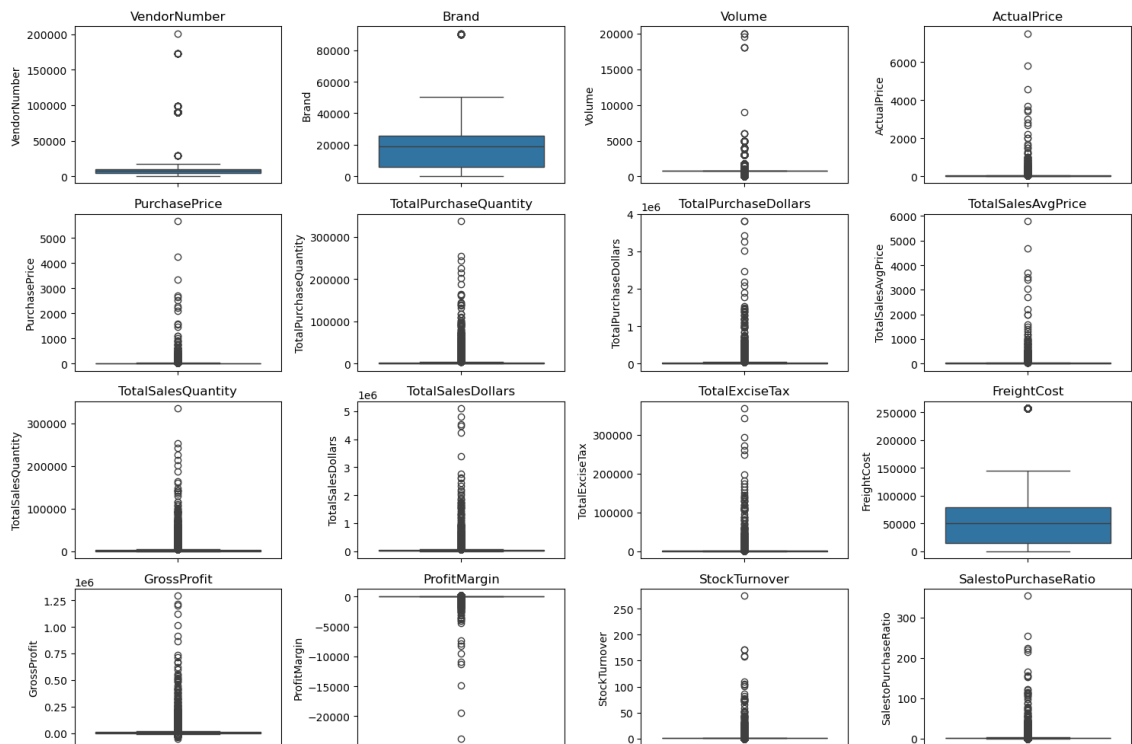


FIGURE 3 Box plots for the same variables, exposing the extreme high-end outliers behind that skew.

Negative & Zero Values

- Gross Profit: Minimum values are **-52,002.78**, indicating losses. Some products or transactions may be selling at a loss due to high costs or selling at discounts lower than the purchase price.
- Profit Margin: Has a minimum **-inf**, which suggests cases where revenue is zero or even lower than costs.
- Total Sales Quantity & Sales Dollars: Minimum values are 0, meaning some products were purchased but never sold. These could be slow-moving or absolute stock.

Outliers Indicated by High Standard Deviation

- Purchase Price & Actual Price: The max values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating potential premium products.
- Freight Cost: Huge variation, from 0.09 to 257,032.07, suggests logistics insufficiencies or bulk shipments.
- Stock Turnover: Ranges from 0 to 274.5, implying some products sell really fast while others remain in stock indefinitely. Value more than 1 indicates that sold quantity for that product is higher than purchased quantity due to either sales are being fulfilled from older stock.

3.2 Data Filtering

To improve the reliability of the insights, the following data points were removed:

Exclusion Criteria

- Excluded records with zero sales because, without time-series data, we cannot determine whether the product is newly introduced or has been available for a while but is not selling.
- Excluded records where total sales quantity is higher than total purchase quantity because some of the sales may have come from older stock, which could make newly added products look less or more successful than they actually are.
- Excluded records with a profit margin of **-101% or lower** because such extreme losses may indicate (may vary business to business):
 - A newly introduced product which has just started selling, or
 - A large amount of unsold inventory, which businesses may simply clear out, discard, or return to the vendor rather than analyse further.

Keeping such records could therefore lead to misleading conclusions about product and vendor performance.

```
df_2 = pd.read_sql("""
SELECT *
FROM vendor_sales_summary
WHERE TotalSalesQuantity > 0
AND NOT (TotalSalesQuantity > TotalPurchaseQuantity)
AND ProfitMargin > -101
""", conn)
```

FIGURE 4 SQL filter excluding zero-sales, stock-inflated, and extreme-loss records.

3.3 Correlation Insights

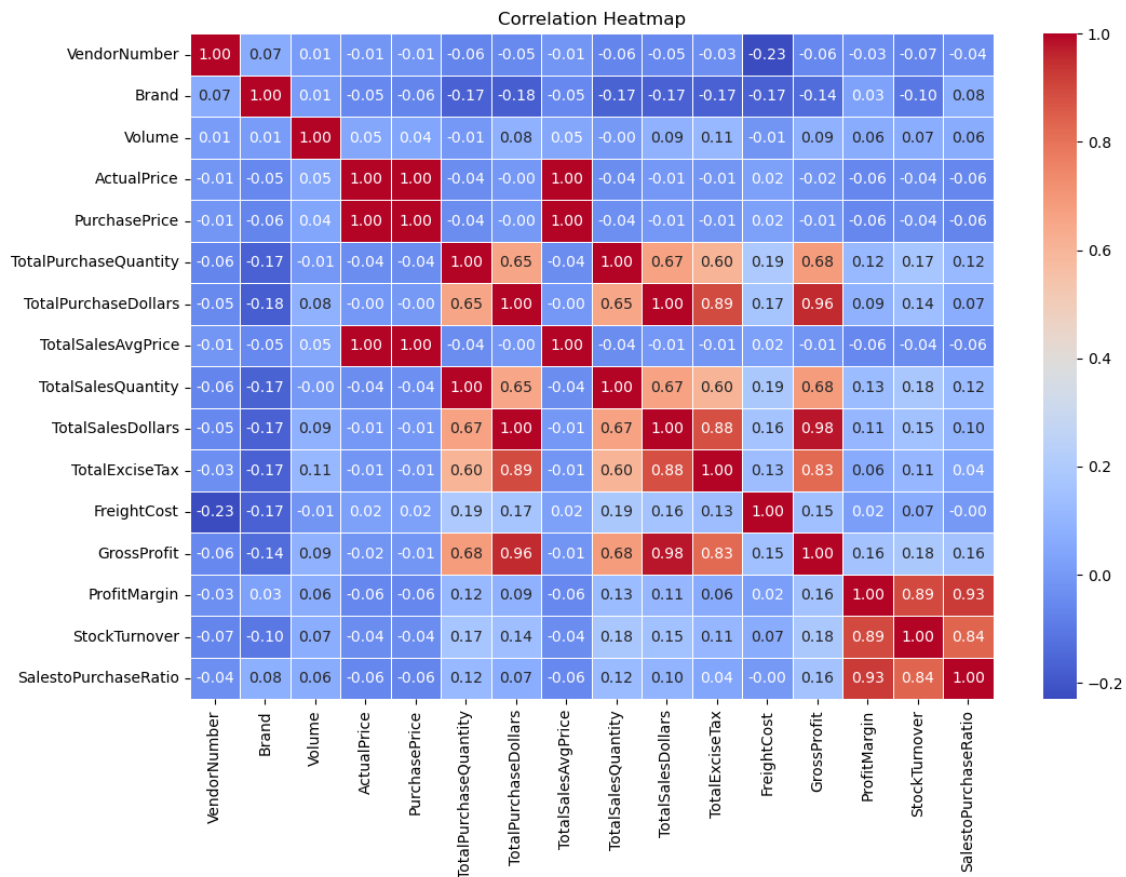


FIGURE 5 Correlation heatmap across all numeric vendor, purchase, and sales metrics.

- PurchasePrice has weak negative correlation with TotalSalesDollars (-0.01) and GrossProfit (-0.01), suggesting that price variations do not significantly impact sales revenue or profit.
- Strong correlation between total purchase quantity and Total Sales Quantity (**1.00**), confirming efficient inventory turnover.
- Weak Negative correlation between profit margin & total sales average price (-0.06) suggests that, for some products as sales price increases, margins decrease, possibly due to competitive pricing pressures.
- Stock Turnover has positive correlations with both GrossProfit (0.18) and ProfitMargin (0.89), indicating that faster turnover necessarily results in higher profitability.
- Total Sales Quantity shows a strong positive correlation with Gross Profit (0.68), indicating that higher sales volumes are generally associated with higher gross profits.

3.4 Research Questions & Key Findings

3.4.1 Brands with Lower Sales but Higher Profit Margins

	Description	TotalSalesDollars	ProfitMargin
4443	Redtree Moscato	8.99	35.48
1198	Ch Lestiac Cotes de Bordeaux	9.49	35.09
4662	Santa Rita Organic Svgn Bl	9.99	66.47
1208	Ch Maycott Red	10.49	35.08
5760	ZIPZ Pnt Grigio	10.74	35.75
...	...	...	...
5543	Viberti Chard	308.23	64.84
3045	Kirkland Signature Cab Svgn	311.76	39.26
3018	Kesselstatt Piesp G Kab Rsl	314.79	35.09
29	2 Gingers Irish Whiskey	321.86	40.89
3211	Laurent Reverdy Sancerre	323.76	35.51

91 rows × 3 columns

FIGURE 6 The identified low-sales, high-margin brands (91 rows), ranked by sales value.

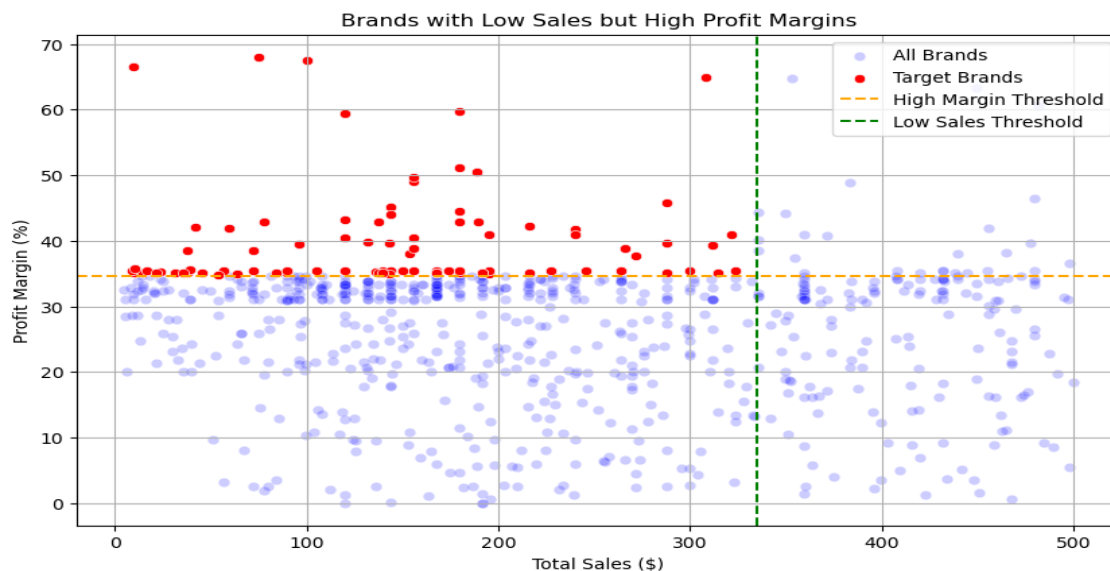


FIGURE 7 Sales value vs. profit margin — the low-sales, high-margin brands highlighted in red.

There are **91 brands** which have low sales but high profit margin. Targeted marketing, promotional campaigns or price adjustments may improve their sales volume without affecting their high profitability.

3.4.2 Brands with Higher Sales but Lower Profit Margins

	Description	TotalSalesDollars	ProfitMargin
4523	Rocky Peak Hard Cinnamon	72075.00	-2.56
4449	Remy Martin 1738 Accord Royl	72486.63	-13.23
5130	Tanteo Jalapeno Tequila	80547.53	-18.22
4028	OuterSpace Vodka	81795.02	-15.12
2780	Jack Daniels Barrel Proof	89296.41	-5.08
2063	Fernet-Branca Italian Liquer	90170.16	-11.39
4965	St Remy VSOP Authentic Brndy	94103.99	-29.21
2028	Fabrizia's Margarita RTD	94406.84	-2.66
1385	Clase Azul Reposado Tequila	95499.46	-17.39
395	Banfi Rosa Regale	96923.38	-5.89
1556	Crown Royal Nrth Harvest Rye	113966.29	-1.53
789	Buehler Znfld Napa	119616.75	-15.68
5515	Vendange Chard	157952.39	-0.88
3061	Knob Creek Rye	162245.21	-18.12
2374	Glen Ellen Cab Svgn RSV	179932.54	-11.18
1370	Ciroc Coconut Vodka	192947.38	-0.95
4214	Pinnacle Tropical Punch	197259.99	-10.22
5030	Stolichnaya Razberi	274277.56	-18.98
2873	Jim Beam Black	296227.06	-23.52
994	Capt Morgan Spiced Rum 100	573230.06	-6.08
5304	Three Olives Vodka	1408861.65	-16.97

21 rows, 3 columns

FIGURE 8 The identified high-sales, low-margin brands (21 rows), ranked by sales value.

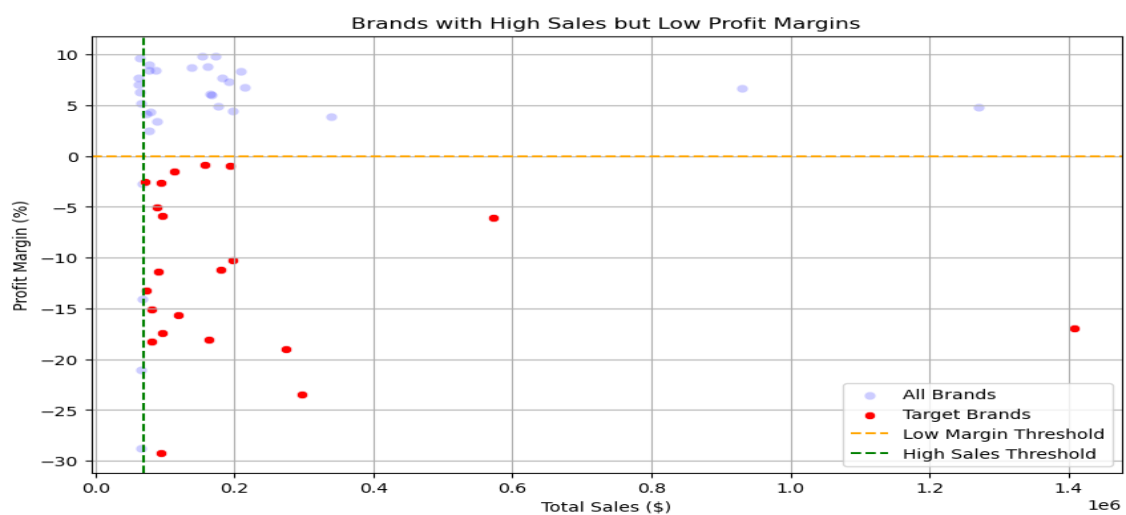


FIGURE 9 Sales value vs. profit margin — the high-sales, low-margin brands highlighted in red.

There are **21 brands** which have high sales but low or negative profit margins. Since these brands have good sales, we should not discontinue them just because they are not profitable. Instead, we should discuss with vendors and look for ways to reduce their purchase prices. This can help improve the profit margins while maintaining the good sales of these brands.

3.4.3 Vendors & Brands with the Highest Sales Performance

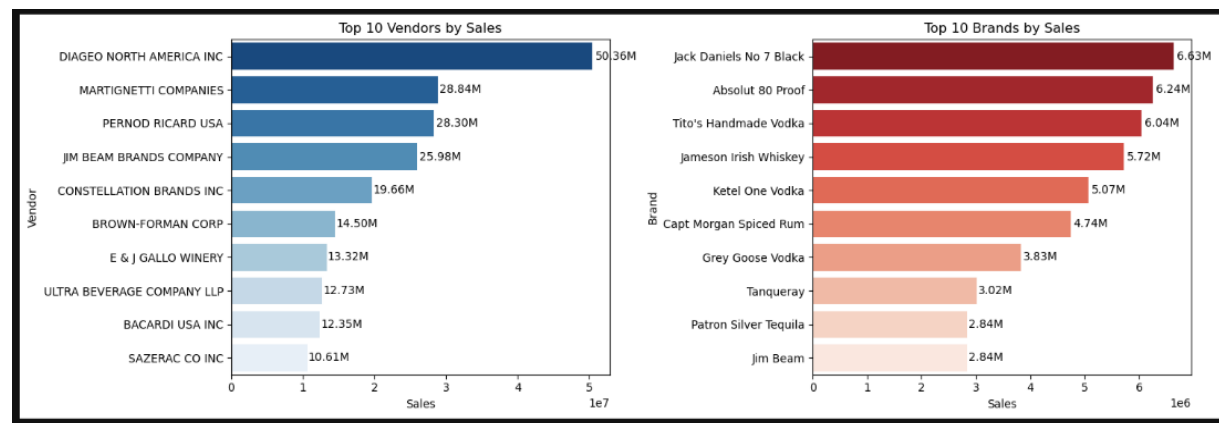


FIGURE 10 Top vendors and brands ranked by total sales value.

Top Vendors

- The business should maintain strong and long-term relationships with these top vendors, as they contribute a major share of the business's total sales. Any disruption or loss of even one of these important vendors could significantly reduce sales and affect overall business performance.

Top Brands

- The business should ensure that these top-selling brands always have sufficient stock to meet customer demand and avoid losing potential sales. At the same time, low-performing brands should be promoted strategically to improve their sales without affecting the availability and sales of these high-performing brands.

3.4.4 Vendors & Brands with the Lowest Sales Performance

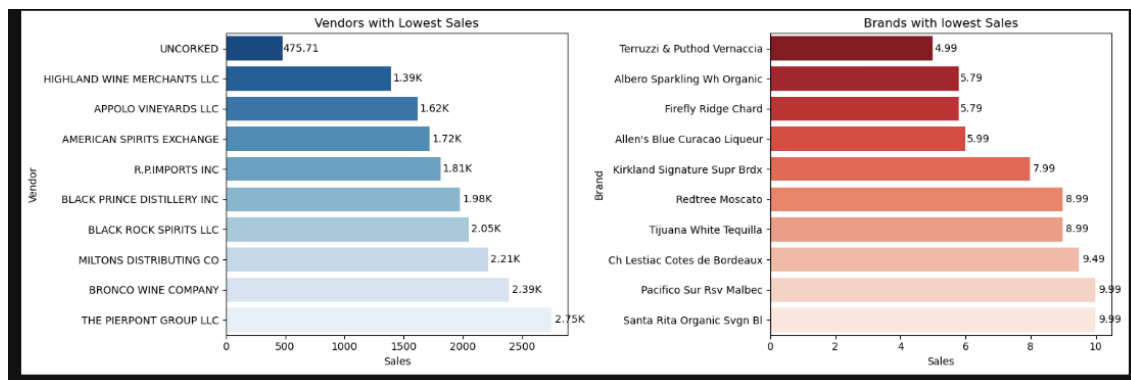


FIGURE 11 Bottom vendors and brands ranked by total sales value.

Low-Sales Vendors

- The business should review these vendors and reduce purchases from them if their products have consistently low demand. This can help avoid investing money in slow-moving products.

Low-Sales Brands

- These brands should be reviewed based on their sales and profitability. If they continue to perform poorly, their stock can be reduced or they can be discontinued, allowing more focus on better-performing brands.

3.4.5 Purchase Concentration Among Top Vendors

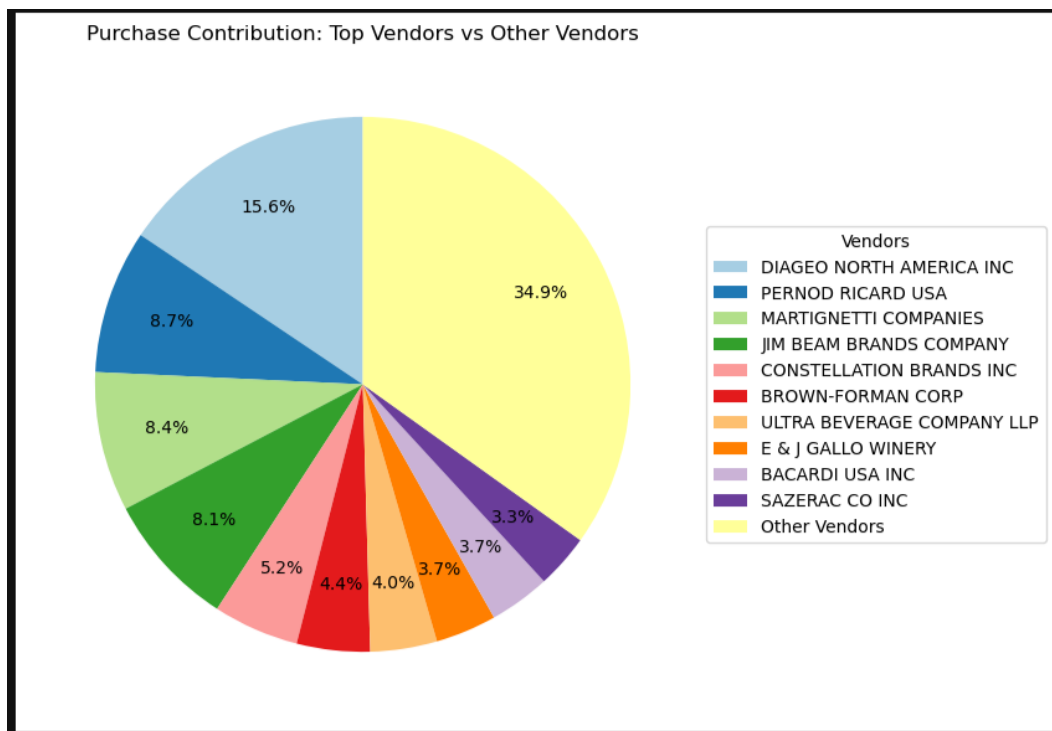


FIGURE 12 Purchase contribution — top 10 vendors vs. all other vendors.

	VendorName	TotalPurchaseDollars	GrossProfit	TotalSalesDollars	PurchaseContribution%	Cumulative_Contribution
25	DIAGEO NORTH AMERICA INC	38.31M	12.06M	50.36M	15.57	15.57
66	PERNOD RICARD USA	21.38M	6.92M	28.30M	8.69	24.27
55	MARTIGNETTI COMPANIES	20.73M	8.10M	28.84M	8.43	32.70
44	JIM BEAM BRANDS COMPANY	20.04M	5.94M	25.98M	8.15	40.84
20	CONSTELLATION BRANDS INC	12.75M	6.92M	19.66M	5.18	46.03
12	BROWN-FORMAN CORP	10.87M	3.63M	14.50M	4.42	50.44
103	ULTRA BEVERAGE COMPANY LLP	9.91M	2.82M	12.73M	4.03	54.48
30	E & J GALLO WINERY	9.06M	4.26M	13.32M	3.68	58.16
7	BACARDI USA INC	9.03M	3.31M	12.35M	3.67	61.83
77	SAZERAC CO INC	8.05M	2.56M	10.61M	3.27	65.10

FIGURE 13 Purchase share by vendor, ranked.

These top 10 vendors contribute around **65.10%** of total purchases, making them a major source of our inventory. The business should maintain strong relationships with these vendors, as losing even one of them could create a significant shortage in inventory. At the same time, the business should gradually increase purchases from other reliable vendors to reduce its dependence on these top 10 vendors and lower the supply risk.

3.4.6 Impact of Bulk Purchasing on Unit Price

```
OrderSize
Small    32.01
Medium   17.37
Large    11.18
Name: UnitPurchasePrice, dtype: float64
```

FIGURE 14 Average unit purchase price by order size.

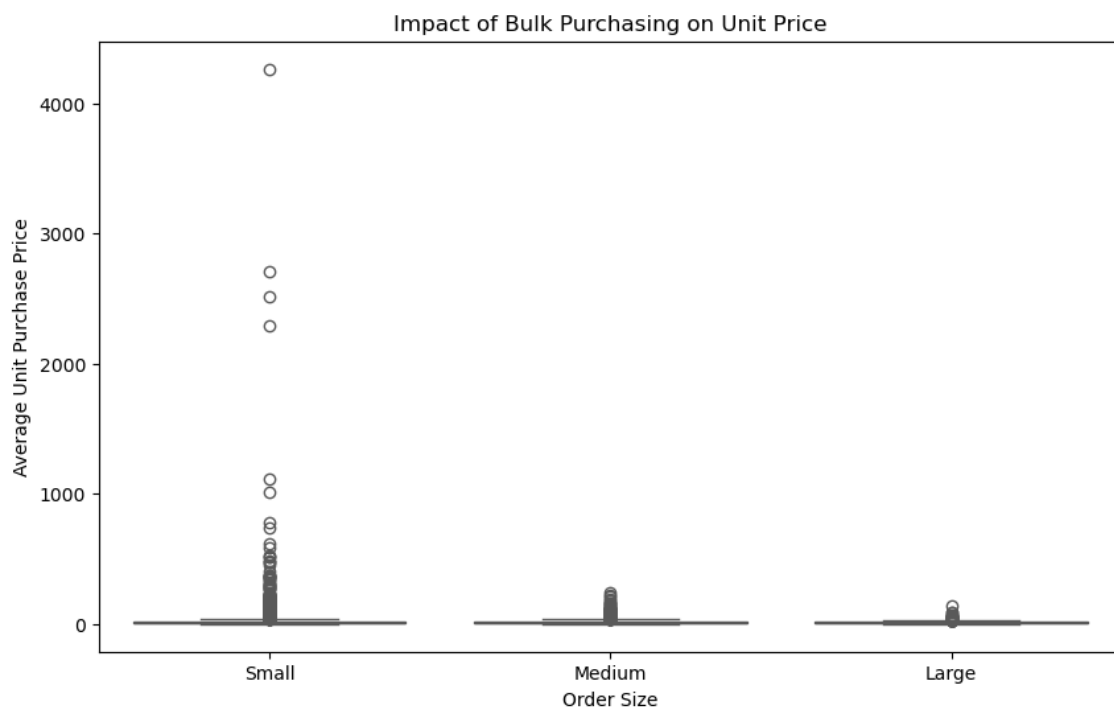


FIGURE 15 Unit purchase price across order-size bands.

- Buying products in bulk (*Large Order Size*) from vendors leads to lowest average unit price (**\$11.18 per unit**), meaning higher margins if business can manage inventory efficiently.
- Larger order sizes result in lower per-unit purchase costs, indicating that bulk purchasing allows the business to achieve better pricing from vendors.

3.4.7 Vendors with the Lowest Inventory Turnover

Top 10 vendors with the lowest Stock Turnover :

	VendorName	StockTurnover
0	UNCORKED	0.36
1	VINEYARD BRANDS LLC	0.44
2	Russian Standard Vodka	0.45
3	GILMANTON WINERY & VINEYARD	0.47
4	APPOLO VINEYARDS LLC	0.47
5	Serralles Usa LLC	0.53
6	SOUTHERN GLAZERS W&S OF NE	0.60
7	NICHE W & S	0.61
8	HIGHLAND WINE MERCHANTS LLC	0.61
9	ALISA CARR BEVERAGES	0.63

FIGURE 16 Vendors ranked by lowest inventory turnover.

3.4.8 Vendors with the Highest Unsold Inventory Value

Total Unsold Capital : 13.21M

FIGURE 17 Total unsold capital tied up across vendors.

Top 10 vendors with the highest Unsold Inventory Value:

	VendorName	UnsoldInventoryValue	UnsoldInventoryContribution%
25	DIAGEO NORTH AMERICA INC	1.60M	12.09
55	MARTIGNETTI COMPANIES	1.53M	11.58
44	JIM BEAM BRANDS COMPANY	1.02M	7.74
103	ULTRA BEVERAGE COMPANY LLP	1.00M	7.58
66	PERNOD RICARD USA	708.17K	5.36
65	PERFECTA WINES	627.36K	4.75
51	M S WALKER INC	529.60K	4.01
113	WILLIAM GRANT & SONS INC	492.70K	3.73
30	E & J GALLO WINERY	458.33K	3.47
20	CONSTELLATION BRANDS INC	431.59K	3.27

The vendor DIAGEO NORTH AMERICA INC holds the highest Unsold Inventory Value of \$1.60M, contributing 12.09% to the total Unsold Inventory Value of \$13.21M.

FIGURE 18 Vendors holding the highest unsold inventory value.

3.4.9 Profit Margin Gap: Low- vs. High-Performing Vendors

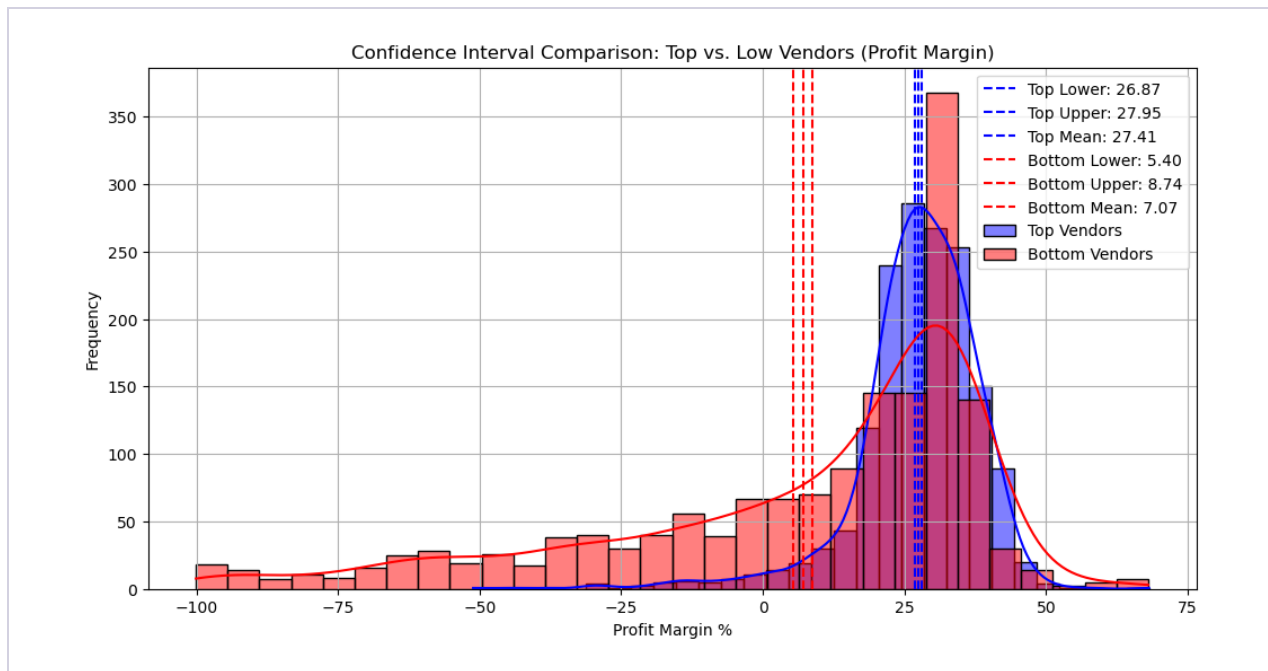


FIGURE 19 Confidence intervals for profit margin — low-performing vs. top-performing vendors.

Interpretation

- The confidence interval for low-performing vendors (**5.40% to 8.74%**) is significantly lower than that of top-performing vendors (**26.87% to 27.95%**). This suggests that sales from low performing vendors may generate low profit margins due to their lower sales volumes, indicating that sales volume leads to higher profitability.
- For High-Performing vendors: We can try to implement some discounts or marketing campaigns to uplift the sales for their products more, which could result in higher Gross Profit as correlation between TotalSalesDollars and GrossProfit is 0.98.
- For Low-Performing vendors: These vendors have low profit margins, because of their lower sales. This could be because of customer interests, quality issues or due to some other factors, the responsible team should investigate into this.

3.4.10 Statistical Validation of the Profit Margin Difference

Hypothesis Test

Null Hypothesis: There is no significance difference in the mean profit margins of top-performing and low-performing vendors.

Alternate Hypothesis: The mean profit margins of top-performing and low-performing vendors are significantly different.

Result: **There is a significant difference in profit margins between top performing vendors and low performing vendors, Reject Null Hypothesis.**

Implication: High-performing vendors may achieve better profit margins due to the larger quantity of products they sell compared to Low-Performing vendors.

SECTION 04

Power BI Dashboard Overview

All insights above, consolidated into two interactive dashboard pages

Every analysis, insight, and visual above has been consolidated into an interactive Power BI dashboard for stakeholders — giving a clear, at-a-glance view of sales, vendor performance, profitability, purchasing, and inventory to support data-driven decisions.

4.1 Primary KPIs

116 TOTAL VENDORS	6K TOTAL BRANDS	\$245.95M TOTAL PURCHASE COST
\$334.57M TOTAL SALES VALUE	15.19% TOTAL PROFIT MARGIN	\$13.21M TOTAL UNSOLD CAPITAL

4.2 Dashboard — Page 1

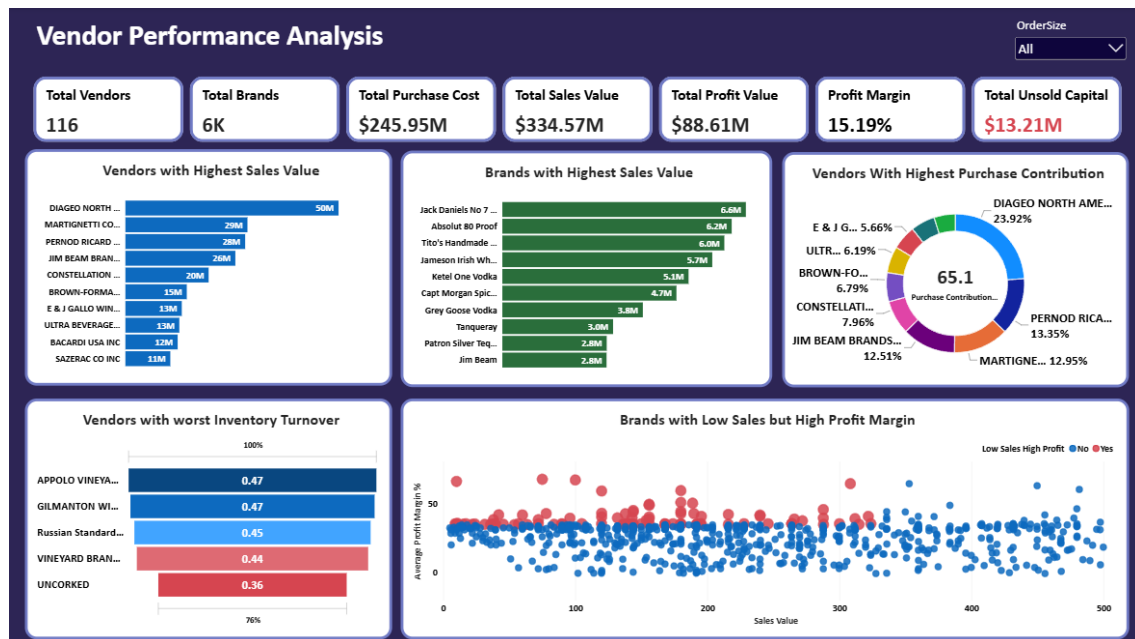


FIGURE 20 Dashboard page 1 — sales, purchase contribution, and inventory-turnover overview.

- Vendors With Highest Sales Value — top 10 vendors by total sales (*bar chart*)
- Brands With Highest Sales Value — top 10 brands by total sales (*bar chart*)
- Vendors With Highest Purchase Contribution — share of the top 10 purchasing vendors (*donut chart*)
- Vendors With Worst Inventory Turnover — 5 vendors with the lowest turnover (*funnel chart*)
- Brands With Low Sales but High Profit Margin — the 91-brand opportunity set (*scatter chart*)

4.3 Dashboard — Page 2

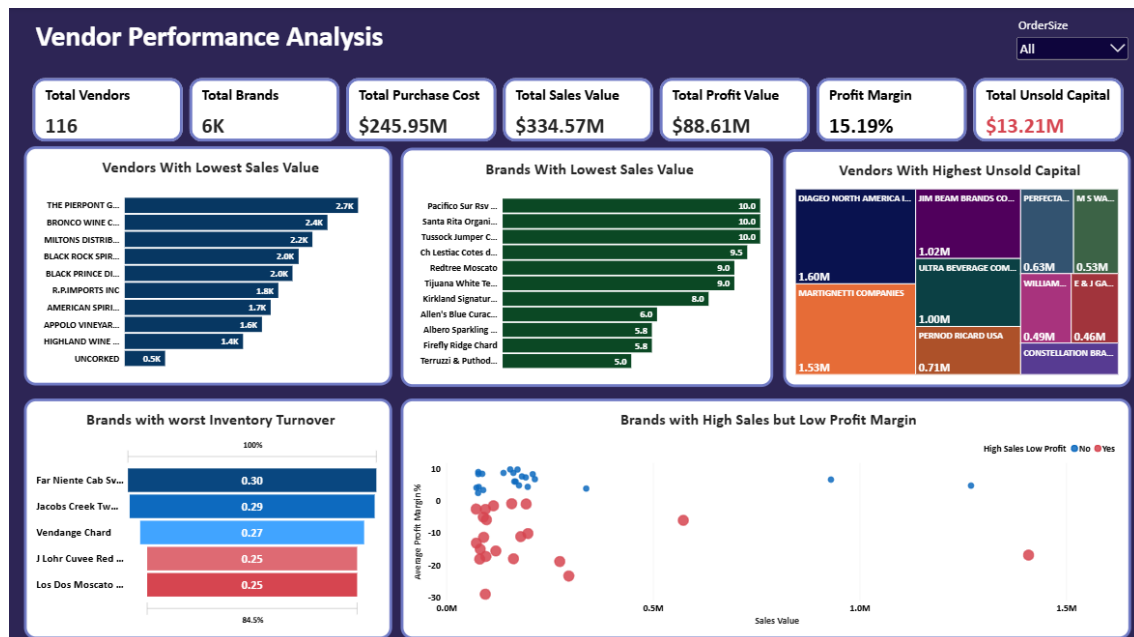


FIGURE 21 Dashboard page 2 — lowest performers, unsold capital, and margin trade-offs.

- Vendors With Lowest Sales Value — bottom 10 vendors by total sales (*bar chart*)
- Brands With Lowest Sales Value — bottom 10 brands by total sales (*bar chart*)
- Vendors With Highest Unsold Capital — 10 vendors holding the most unsold value (*treemap*)
- Brands With Worst Inventory Turnover — 5 brands with the lowest turnover (*funnel chart*)
- Brands With High Sales but Low Profit Margin — the 21-brand renegotiation set (*scatter chart*)

SECTION 05

Final Recommendations

Turning the findings above into concrete next steps

- 01** Maintain strong relationships with top vendors, as they supply the majority of purchases and inventory.
- 02** Negotiate lower purchase prices for high-selling, low-profit brands to improve their margins.
- 03** Keep sufficient stock of top-selling brands so that we don't lose sales due to shortages.
- 04** Reduce purchases of slow-moving products to limit capital tied up in unsold inventory.
- 05** Use bulk purchasing where possible to secure lower unit prices from vendors.
- 06** Reduce unsold inventory through discounts, promotions, or vendor returns.
- 07** Promote low-selling, high-profit brands to grow sales without diluting profitability.
- 08** Increase purchases from other reliable vendors to reduce dependence on the top 10.
- 09** Improve inventory turnover by focusing on products that sell slowly and finding ways to increase their sales.